



Introduction to Bioengineering Theory (BBL1020; 3-0-0)

Lecture-8: 2 Classes



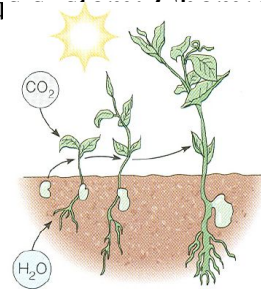
*Please Listen Carefully
&*

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Thermodynamics and Energy Coupling

Biological Thermodynamics

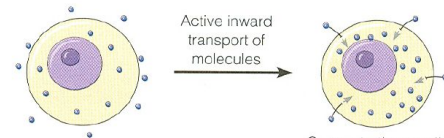
- All chemical reactions in cells involve energy transformations. The energy can be transduced into new forms. Thermodynamics is the study of energy in its various forms and the conversion of one form of energy to another.
- Thermodynamics is concerned with energy storage, transformation and dissipation. Cells store energy, transform and dissipate to drive unfavorable reactions.
- In living cells, thermodynamic changes are essential for biological functions such as growth, reproduction, photosynthesis and respiration. Light to Chemical: photosynthesis; Chemical to Chemical: cellular respiration; Chemical to Electrical: Nervous system; Chemical to Mechanical: Muscle contraction



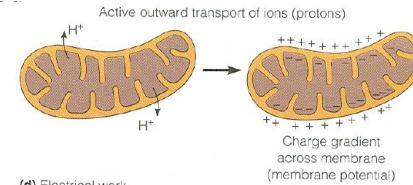
(a) Synthetic work



(b) Mechanical work



(c) Concentration work



(d) Electrical work



(e) Heat



(f) Bioluminescent work

Types of Systems

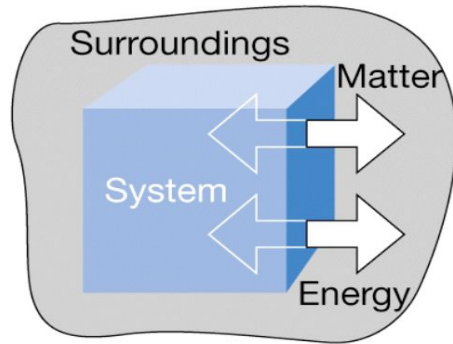


□ Systems are classified according to the nature of their boundary against their environment:

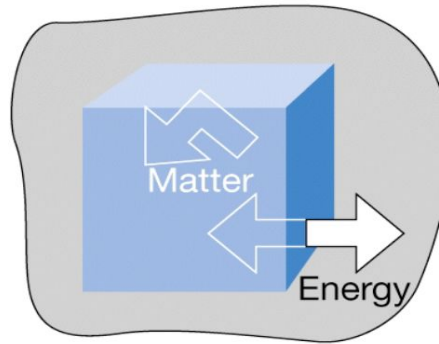
Isolated system: idealized system that does not exchange any kind of energy or matter with its environment.

Closed system: this system can exchange all kinds of energy with its environment, but not matter.

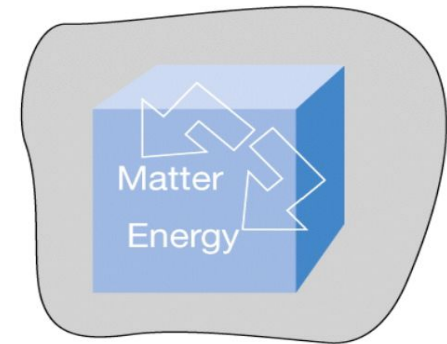
Open system: it can exchange both energy and matter with its environment.



(a) Open



(b) Closed



(c) Isolated

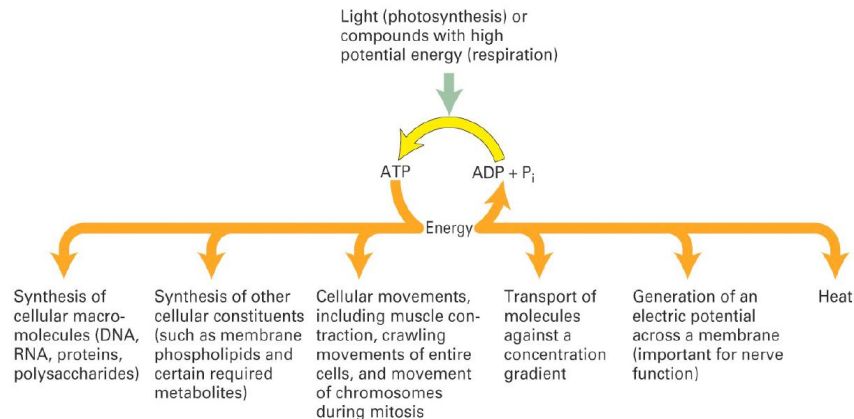
Laws of Thermodynamics



The Energy is conserved

The total energy of a system and its surroundings is constant

In any physical or chemical change, the total amount of energy in the universe remains constant, although **the form of the energy may change**.



The Second Law of Thermodynamics

- During every energy transfer or transformation, some energy is unusable, and is often lost as heat
- According to the **second law of thermodynamics**:
 - Every energy transfer or transformation increases the **entropy** (disorder) of the universe

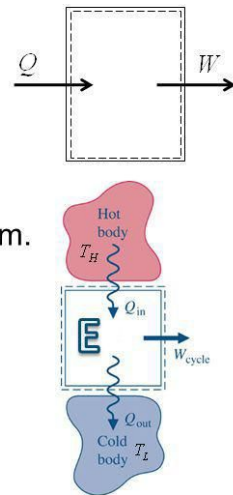
First, consider a closed system. For this system, we know that the Second Law is,

$$S_T + S_P = S_G \rightarrow \therefore dS_T + dS_P = dS_G$$

Recall that Clausius discovered entropy analyzing a Carnot cycle as a closed system. His discovery was,

$$dS = \left(\frac{\bar{d}Q}{T} \right)_{rev}$$

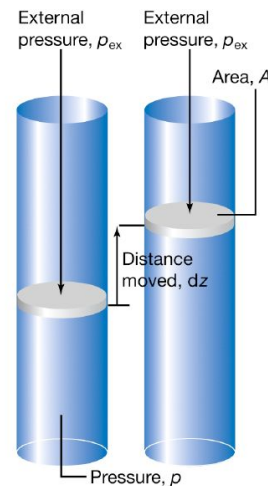
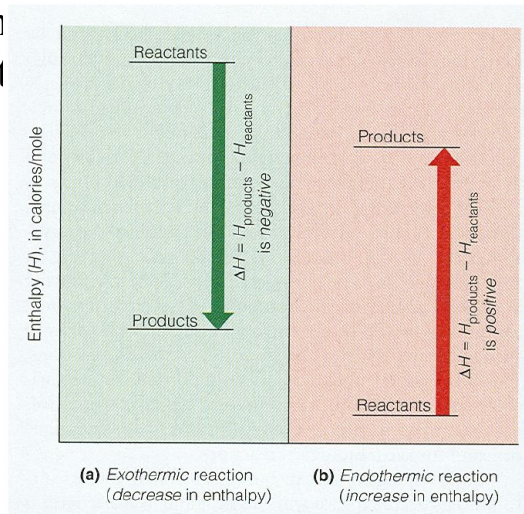
T in this expression is the *boundary temperature* where Q is being transferred.



Gibbs Free Energy, Entropy and Enthalpy



- Free energy refers to the amount of energy available during a chemical reaction to perform cellular functions. The Gibbs free energy is a thermodynamic quantity which can determine whether a reaction is spontaneous or not.
- Gibbs free energy equation: $\Delta G = \Delta H - T\Delta S$; where ΔG is Gibbs free energy in KJ, ΔH is enthalpy change, ΔS in entropy change and T is temperature in Kelvin.
- ΔH is the drive towards stability; ΔS is the drive towards disorder; ΔG is the net driving force of a chemical reaction.
- ΔG value depend on temperature, pressure and the concentrations of reactants and products.
- If $\Delta G < 0$, the reaction is spontaneous; If $\Delta G = 0$, the reaction is non-spontaneous; If $\Delta G > 0$, the reaction is non-spontaneous.



- ΔH during a chemical reaction is the heat absorbed or released in the breaking and formation of bonds.

Gibbs Free Energy, Entropy and Enthalpy



The Gibbs free energy (ΔG)

$$\Delta S_{\text{total}} = \Delta S_{\text{system}} + \Delta S_{\text{surroundings}}$$

$$\Delta S_{\text{surroundings}} = -\Delta H_{\text{system}}/T$$

$$\Delta S_{\text{total}} = \Delta S_{\text{system}} - \Delta H_{\text{system}}/T$$

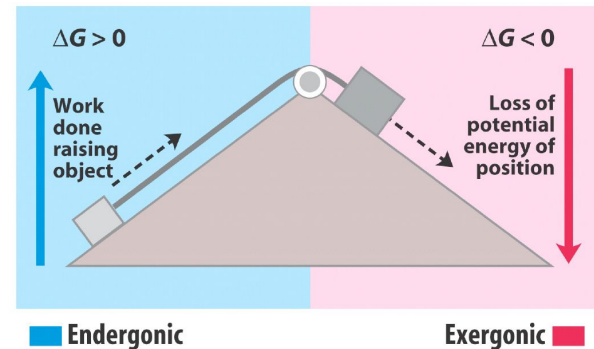
$$-T\Delta S_{\text{total}} = \Delta H_{\text{system}} - T\Delta S_{\text{system}}$$

$$\Delta G = \Delta H_{\text{system}} - T\Delta S_{\text{system}}$$

For a reaction to be spontaneous, the entropy of the universe, ΔS_{total} , must increase

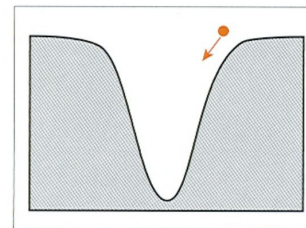
$$\Delta S_{\text{system}} > \Delta H_{\text{system}}/T \quad \text{or} \quad \Delta G = \Delta H_{\text{system}} - T\Delta S_{\text{system}} < 0$$

The **free energy** must be **negative** for a reaction to be **spontaneous**!



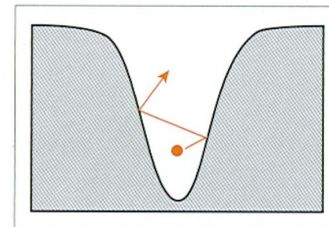
Energy coupling links reactions

Enthalpy



Things tend to want to roll downhill: ΔH tends to be **negative**

Entropy



Thermal (Brownian) motion tends to kick things uphill: ΔS tends to be **positive**

The **Enthalpic** term

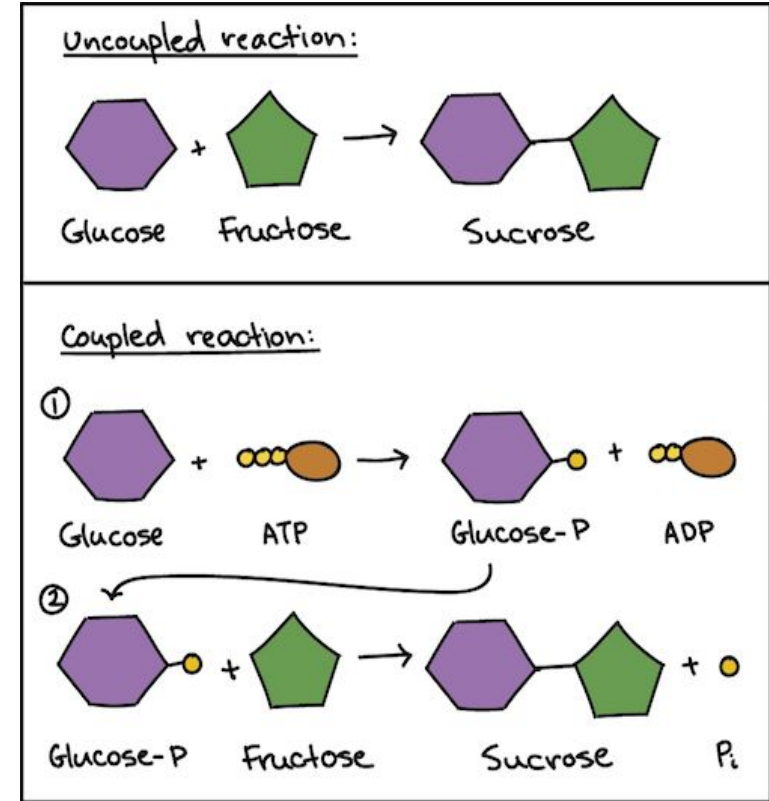
- Changes in bonding
- van der Waals
- Hydrogen bonding
- Charge interactions

The **Entropic** term

- Changes the arrangement of the solvent or counterions
- Reflects the degrees of freedom
- Rotational & Translational changes

Energy Coupling

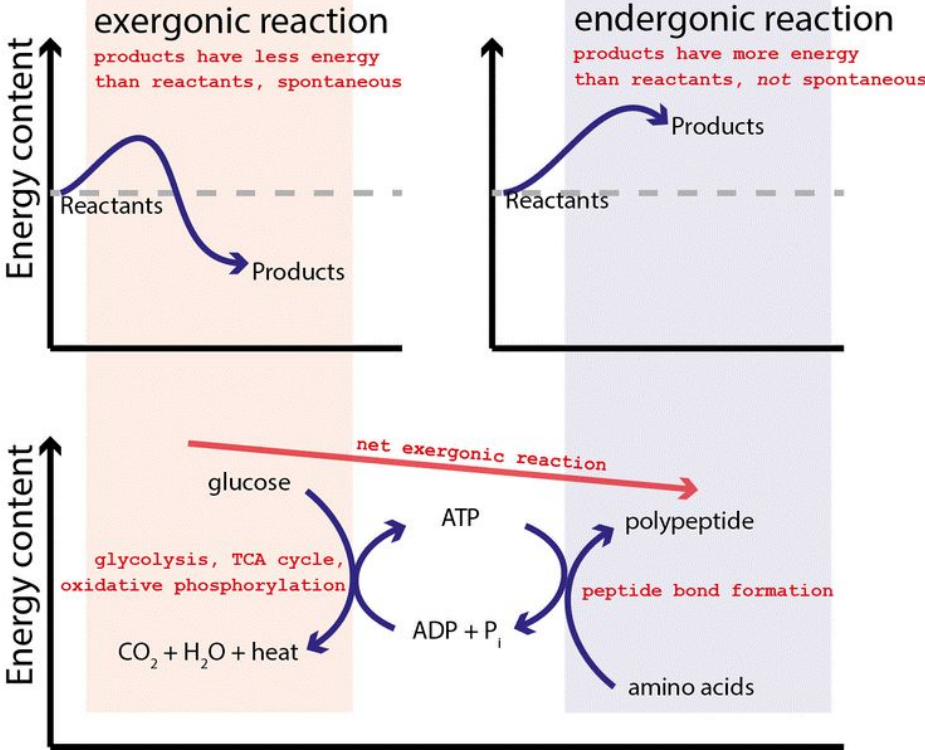
- **Energy coupling**, by definition, stands for a concept of coupling two biological reactions; meaning energy generated from one reaction is used to drive the second reaction. Two different reactions or biological systems are coupled together or put into synchrony this way.
- All cells have majorly 2 types of reactions going on in them: (1) **exergonic reactions** that are energy-releasing reactions (thus, energetically-favorable) and (2) **endergonic reactions** that are the energy-demanding reactions (thus, energetically-unfavorable).
- Many times, the [cell](#) couples these two reactions and bridges the gap between them. The [energy](#) that is released by the exergonic reactions is channeled down to the endergonic reactions to make them energetically-favorable.



Spontaneous & Non-spontaneous reactions



Character	Spontaneous Reaction	Non-Spontaneous Reaction
Energy-wise	Exergonic	Endergonic
ΔG	$\Delta G > 0$	$\Delta G < 0$
Thermodynamically	Favorable	Unfavorable
Functioning	Drives the thermodynamically unfavorable reaction	Require energy application externally

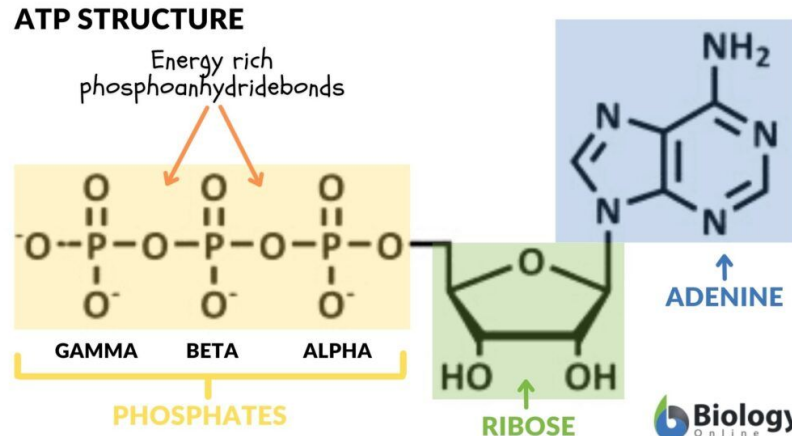


Energy Coupling



- In biological organisms, ATP is the source of energy for all cellular work. This smart work by the cell ensures that the cellular reactions never run out of the energy source i.e., ATP.
- This “*reactions linking or coupling*” is a vital mechanism that ensures that the cellular machinery never grinds down to a halt and the cell remains alive! *It is through this energy coupling that cells use ATP in the endergonic reactions...*

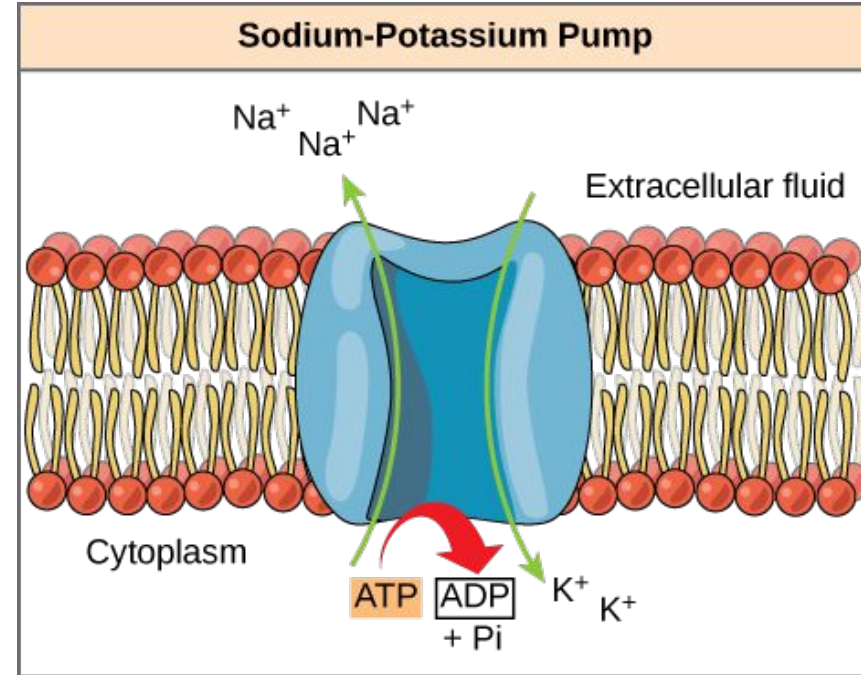
ATP stands for adenosine triphosphate. It is a small molecule that's commonly called the energy currency of the cell. The basic structure of ATP has 3 main components. They are adenine (the nitrogenous base), ribose (the sugar) and three phosphate groups bonded one after another (labeled as *alpha*, *beta*, *gamma* from point of attachment to end).



Energy Coupling in Na-K pumps

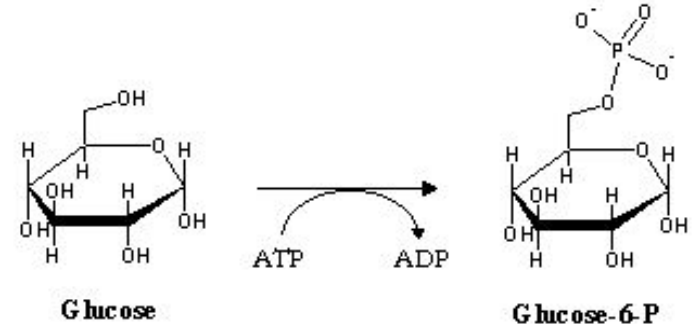
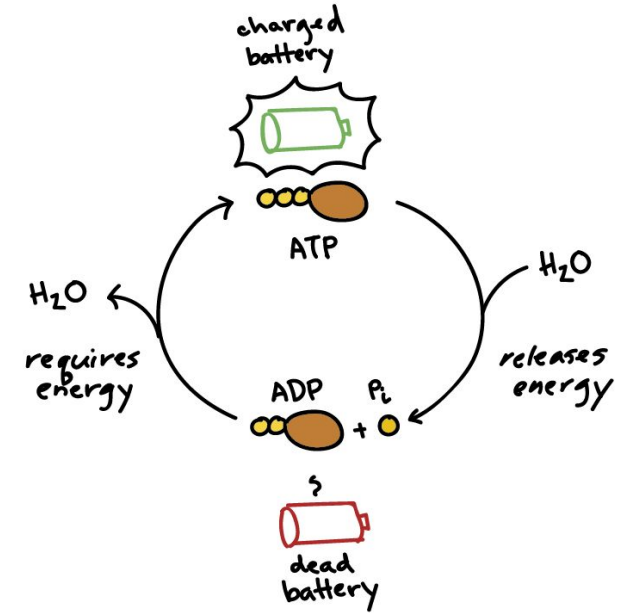


- Cells couple the exergonic reaction of ATP hydrolysis with the endergonic reactions (energy is absorbed from surroundings) of cellular processes. For example, transmembrane ion pumps in nerve cells use the energy from ATP to pump ions across the cell membrane and generate an action potential.
- The **sodium-potassium pump (Na^+/K^+ pump)** drives **sodium out of the cell and potassium into the cell**. When ATP is hydrolyzed, it transfers its **gamma phosphate** to the pump protein in a process called **phosphorylation**. The Na^+/K^+ pump gains the free energy and **undergoes a conformational change**, allowing it to release **three Na^+** to the outside of the cell.
- Two extracellular K^+** ions bind to the protein, causing the protein to change shape again and discharge the phosphate. By donating free energy to the Na^+/K^+ pump, phosphorylation drives the endergonic reaction.



Energy Coupling in metabolism

- During cellular metabolic reactions, or the synthesis and breakdown of nutrients, certain molecules must be altered in their conformation to become substrates for the next step in the reaction series.
- In the very first steps of cellular respiration, **glucose is broken down through the process of glycolysis**. ATP is required for the **phosphorylation of glucose**, creating a high-energy but unstable intermediate. This phosphorylation reaction causes a conformational change that allows enzymes to convert the **phosphorylated glucose molecule to the phosphorylated sugar fructose**.
- Fructose is a necessary intermediate for glycolysis to move forward. In this example, the exergonic reaction of ATP hydrolysis is coupled with the endergonic reaction of converting glucose for use in the metabolic pathway.



Other examples of energy coupling

- ATP hydrolysis coupled with **Starch/Glycogen/Cellulose** formation (formation of chains of sugars).
- ATP hydrolysis coupled with the **formation of RNA**.
- ATP hydrolysis coupled with the **synthesis of biomolecular polymers- carbohydrates, lipids, proteins, and nucleic acids**.
- ATP hydrolysis is coupled with important biological pathways like **glycolysis, Krebs cycle, gluconeogenesis**, etc.

